

## 1. Handling Variable Trace Gases ( $\text{CO}$ and $\text{CH}_4$ )

While fixed gases use a baseline 10-term polynomial mapping purely temperature deviations, variable trace gases like Carbon Monoxide ( $\text{CO}$ ), Methane ( $\text{CH}_4$ ), and Nitrous Oxide ( $\text{N}_2\text{O}$ ) require an advanced **7-to-11 term expansion** framework.

Because trace gas concentrations fluctuate independently across different atmospheric layers, their predictors must track the physical abundance of the molecules alongside kinetic thermal changes. SARTA sets up these specialized predictors using the actual layer gas column density ( $q$ ) against a standard reference profile abundance ( $q_{\text{ref}}$ ): [1]

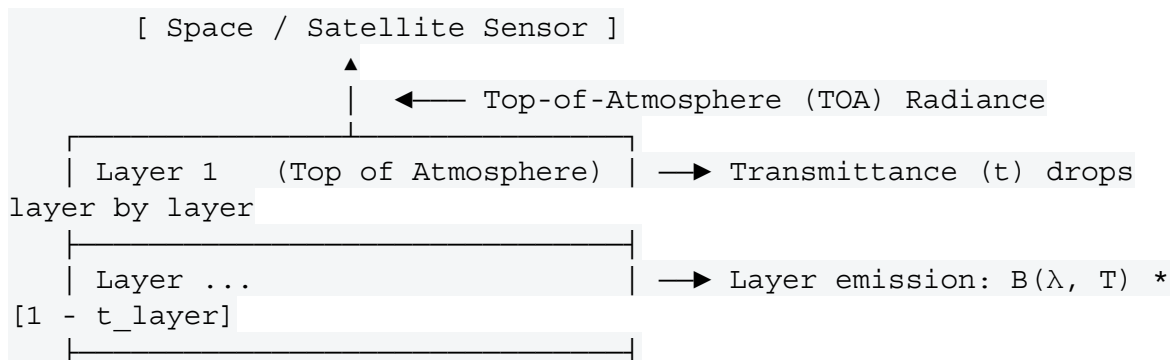
- **Abundance Scaling:** The ratio  $q/q_{\text{ref}}$  is used to scale baseline optical depth up or down depending on the local concentration.
- **Path Concentration Interaction:** The term  $(q/q_{\text{ref}}) \cdot \sec(\theta)$  tracks how a slanted satellite viewing angle increases the effective number of trace gas molecules the sensor senses through that layer.
- **Thermal-Abundance Pairing:** The multi-variable term  $\Delta T \cdot (q/q_{\text{ref}})$  models how temperature variations alter the quantum vibrational-rotational absorption line intensities of that specific trace gas.

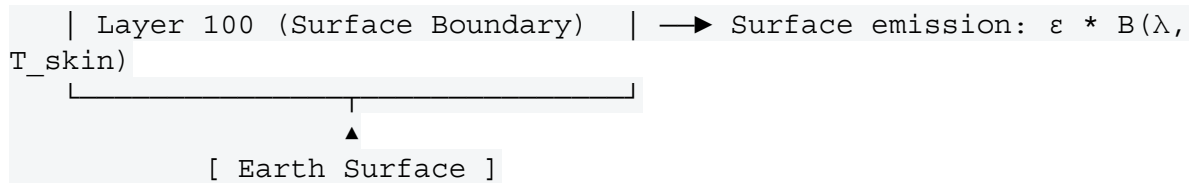
By processing these mixed terms, SARTA isolates individual gas spectral signatures (such as the  $4.67 \mu\text{m}$  carbon monoxide absorption band or the  $7.6 \mu\text{m}$  methane absorption band) without re-running an entire line-by-line model setup.

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## 2. From Layer Optical Depths to Top-of-Atmosphere (TOA) Radiance

Once SARTA computes the individual layer optical depths ( $\tau$ ) for every gas across all 100 layers using the regression polynomials, it executes a vertical integration to solve the **Schwarzschild Radiative Transfer Equation**. This is what translates molecular absorption into the exact energy structure detected by a satellite sensor.





## Step A: Layer-to-Space Transmittance (\$t\_L\$)

First, SARTA sums the individual optical depths of all active gases (fixed, water vapor, ozone, and trace gases) inside a specific layer (\$L\$) to find the total layer optical depth (\$\tau\_{\text{total}}\$). It converts this total into a layer-to-space **transmittance profile** (\$t\_L\$) using Beer's Law along the slanted path:

$$t_L = \exp(-\tau_{\text{total}} \cdot \sec(\theta))$$

## Step B: Numerical Thermal Integration

Working from the surface up to layer 1, SARTA accumulates the total upwelling infrared energy. For a clear sky, the baseline thermal stream consists of two primary parts:

1. **Surface Boundary Emission:** The ground radiates energy upward based on its skin temperature (\$T\_{\text{skin}}\$) and spectral emissivity (\$\epsilon\$), multiplied by the total atmospheric transmittance from the surface to space:  

$$I_{\text{surface}} = \epsilon \cdot B(\lambda, T_{\text{skin}}) \cdot t_{\text{surface}}$$
*(Where \$B\$ represents the fundamental thermal Planck function) [2]*
2. **Atmospheric Layer Emission:** Each individual atmospheric layer emits its own thermal radiation relative to its kinetic layer temperature (\$T\_L\$). This emission is weighted by the layer's internal absorptivity (\$1 - t\_{\text{layer}}\$) and then attenuated by the remaining upper atmosphere as it travels out to space.

SARTA sums these layers numerically to output the final **Top-of-Atmosphere (TOA) Spectral Radiance** (\$I\_{\text{TOA}}\$).

## 3. The Final Step: Converting Radiance to Brightness Temperature (\$T\_B\$)

Satellites measure physical raw energy (Radiance, \$W/m^2/sr/cm^{-1}\$), but meteorologists and data assimilation systems need a clear thermodynamic property.

To make this conversion, SARTA takes the integrated \$I\_{\text{TOA}}\$ value and **inverts the Planck function**. By solving for temperature mathematically, it extracts the final **Brightness Temperature (\$T\_B\$)** in Kelvin: [2]

$$T_B = \frac{c_2 \cdot \nu}{\ln\left(1 + \frac{c_1 \cdot \nu^3}{I_{\text{TOA}}}\right)}$$

*(Where \$\nu\$ is the specific wavenumber of the channel, and \$c\_1, c\_2\$ are the standard*

*physical radiation constants*) [2]

This  $T_B$  represents the equivalent temperature of a perfect blackbody that would emit the exact amount of radiance the satellite channel detects.

Would you like to look at how SARTA calculates its **analytical Jacobians (temperature/water vapor sensitivities)** during this integration process, or explore how it handles **surface reflection** features?

[1] <https://gml.noaa.gov>

[2] <https://cimss.ssec.wisc.edu>